

Mathematics

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Chapter 1

Algebra

1.1 Groups

1.1.1 Introduction to groups

Magmas and properties

Definition 1.1. A *magma* is a pair $(M, *)$ with M a set and

$$* : M \times M \rightarrow M$$

an internal composition law.

Definition 1.2. A magma $(M, *)$ is said to be *associative* if $\forall a, b, c \in M, (a * b) * c = a * (b * c)$.

Definition 1.3. A magma $(M, *)$ is said to be *commutative* if $\forall a, b \in M, a * b = b * a$.

Definition 1.4. An *identity element* of a given magma $(M, *)$ is an element $e \in M$, such that $\forall a \in M, e * a = a * e = a$.

Theorem 1.5. If a magma $(M, *)$ has an identity element, then it is unique.

Proof. If $e, e' \in M$ are identity elements for $*$, then $e = e * e' = e'$. □

Definition 1.6. Given a magma $(M, *)$ with an identity element $e \in M$. An element $a \in M$ is said to be *invertible* if there exists $b \in M$ such that $a * b = b * a = e$. b is then called an inverse of a .

Products of magmas

Definition 1.7. Given two magmas $(M, *)$ and $(N, \bullet)$, we define a canonical structure of magma on the set $M \times N$, defined by the following internal composition law:

$$((a, b), (a', b')) \mapsto (a * a', b \bullet b')$$

Proposition 1.8. *Let M and N be two magmas.*

- (i) *If M and N are associative (resp. commutative), then $M \times N$ is associative (resp. commutative),*
- (ii) *If M and N have identity elements, then $M \times N$ has an identity element.*

Proof. Admitted. □

Proposition 1.9. *Let M and N be two magmas with identity elements. If $a \in M$ and $b \in N$ are invertible, then $(a, b) \in M \times N$ is invertible.*

Proof. Admitted. □

Remark 1.10. The previous construction can be extended to any product of magmas. The propositions (1.8) and (1.9) then remain true.

Morphism of magmas

Definition 1.11. *A morphism of magmas between $(M, *)$ and $(N, \bullet)$ is a map $f : M \rightarrow N$ such that*

$$\forall a, b \in M, f(a * b) = f(a) \bullet f(b)$$

Proposition 1.12. *A composition of morphisms of magmas is a morphism of magmas.*

Proof. Admitted. □

Monoids and properties

Definition 1.13. *A monoid is an associative magma with an identity element.*

Theorem 1.14. *If $(M, *)$ is a monoid, then every invertible element $a \in M$ has a unique inverse, denoted by a^{-1} .*

Proof. Let $b, b' \in M$ be two inverses of a . Then we have

$$b = b * (a * b') = (b * a) * b' = b'.$$

□

Notation 1.15. If $(M, *)$ is a monoid, then for all $a \in M$, we denote:

$$a^0 = e \quad \text{and} \quad a^n = \underbrace{a * \cdots * a}_{n \text{ times}} \quad (\text{for } n \geq 1).$$

Proposition 1.16. *Given a monoid $(M, *)$, and $a, x, y \in M$,*

- (i) *If $a * x = a * y$ and a is invertible, then $x = y$,*

(ii) If x is invertible, then x^{-1} is invertible and $(x^{-1})^{-1} = x$.

(iii) If x and y are invertible, then $x * y$ is also invertible and $(x * y)^{-1} = y^{-1} * x^{-1}$,

(iv) If x is invertible, then for all $n \in \mathbb{N}$, x^n is also invertible and $(x^n)^{-1} = (x^{-1})^n$. Hence we can denote $x^{-n} = (x^n)^{-1}$,

Proof. (i) $x = (a^{-1} * a) * x = a^{-1} * (a * x) = a^{-1} * (a * y) = (a^{-1} * a) * y = y$.

(ii) Direct consequence of the invertible element definition.

(iii) $(x * y) * (y^{-1} * x^{-1}) = x * y * y^{-1} * x^{-1} = e$, and similarly for the other way.

(iv) It suffices to combine the two preceding properties in a proof by induction. □

Proposition 1.17. *Any product of monoids is a monoid.*

Proof. It is induced by (1.8). □

Definition 1.18. A *morphism of monoids* is a morphism between underlying magmas, such that it preserves the identity element.

Groups

Definition 1.19. A *group* is a monoid such that every element is invertible.

Definition 1.20. The order of a group G , denoted by $|G|$ or $\#G$, is the cardinal of its underlying set.

Definition 1.21. A group is *commutative* or *abelian* if the underlying magma is commutative.

Proposition 1.22. *Any product of groups is a group.*

Proof. It is induced by (1.17) and (1.9). □

1.1.2 Subgroups, morphism of groups

Subgroups

Definition 1.23. Let $(G, *)$ be a group. A subset $H \subseteq G$ is said to be a *subgroup* of G if:

(i) The identity element $e \in H$,

(ii) H is closed under the composition law, i.e., $\forall x, y \in H, x * y \in H$,

(iii) H is closed under taking inverses, i.e., $\forall x \in H, x^{-1} \in H$.

Proposition 1.24. *Let $(G, *)$ be a group. A non-empty subset $H \subseteq G$ is a subgroup of G if and only if:*

$$\forall x, y \in H, \quad x * y^{-1} \in H.$$

Proof. Suppose H is a subgroup of G . Let $x, y \in H$. By condition (iii) of Definition 1.23, $y^{-1} \in H$. Then, by condition (ii), $x * y^{-1} \in H$.

Conversely, suppose $H \neq \emptyset$ and satisfies the given criterion. Since H is non-empty, there exists some $x \in H$. Applying the criterion to x and x yields $x * x^{-1} = e \in H$, which satisfies condition (i). Now, for any $y \in H$, applying the criterion to e and y gives $e * y^{-1} = y^{-1} \in H$, satisfying condition (iii). Finally, for any $x, y \in H$, we know $y^{-1} \in H$, so applying the criterion to x and y^{-1} gives $x * (y^{-1})^{-1} = x * y \in H$, satisfying condition (ii). $\square$

Proposition 1.25. *Given a group G , any intersection of subgroups of G is a subgroup of G .*

Proof. Admitted. $\square$

Morphism of groups

Definition 1.26. *A morphism of groups is a morphism of underlying magmas.*

Definition 1.27. Let $f : G \rightarrow G'$ be a morphism of groups. We say that G is the *domain* of f , and G' is the *codomain* of f .

Proposition 1.28. *Let G, G' be two groups and $f : G \rightarrow G'$ be a morphism of groups. Then $f(e_G) = e_{G'}$ and for all $x \in G$, $f(x^{-1}) = f(x)^{-1}$.*

Proof. Admitted. $\square$

Proposition 1.29. *A composition of morphisms of groups is a morphism of groups.*

Proof. Induced by (1.12). $\square$

Theorem 1.30. *Let G, G' be two groups and $f : G \rightarrow G'$ be a morphism of groups.*

- (i) *If H is a subgroup of G , then $f(H)$ is a subgroup of G' ,*
- (ii) *If H' is a subgroup of G' , then $f^{-1}(H')$ is a subgroup of G .*

Proof. Admitted. $\square$

Definition 1.31. The *kernel* of a morphism of groups $f : G \rightarrow G'$ is defined by

$$\text{Ker}(f) = \{a \in G \mid f(a) = e_{G'}\}$$

Proposition 1.32. *The kernel of a morphism of groups $f : G \rightarrow G'$ is a subgroup of G .*

Proof. Induced by (1.30)-(ii). $\square$

Proposition 1.33. *A morphism of groups $f : G \rightarrow G'$ is injective if and only if its kernel is the trivial subgroup $\{e_G\}$.*

Proof. The if condition: Let $x, x' \in G$ be such that $f(x) = f(x')$, which is equivalent to $f(x)f(x')^{-1} = e_{G'}$. However, $f(x)f(x')^{-1} = f(x)f((x')^{-1}) = f(x(x')^{-1})$. Which implies, by hypothesis $x(x')^{-1} = e_G$. So $x = x'$. The reverse is obvious. $\square$

Definition 1.34. The *image* of a morphism of groups $f : G \rightarrow G'$ is defined by

$$\text{Im}(f) = \{b \in G' \mid \exists a \in G, f(a) = b\}$$

Proposition 1.35. *The image of a morphism of groups $f : G \rightarrow G'$ is a subgroup of G' .*

Proof. Induced by (1.30)-(i). $\square$

Isomorphism of groups

Definition 1.36. An *isomorphism of groups* is a bijective morphism of groups.

Notation 1.37. If $f : G \rightarrow G'$ is an isomorphism, then we say that G is isomorphic to G' . If $G = G'$, we say that f is an automorphism.

Theorem 1.38. *The composition of two isomorphisms of groups is an isomorphism of groups.*

Proof. Obvious. $\square$

Theorem 1.39. *If f is an isomorphism of groups, then the inverse f^{-1} is a morphism of groups. Moreover, it is an isomorphism of groups.*

Proof. Let $f : G \rightarrow G'$ be an isomorphism of groups and $y, y' \in G'$. Then

$$\begin{aligned} f^{-1}(yy') &= f^{-1}(f(f^{-1}(y))f(f^{-1}(y'))) \\ &= f^{-1}(f(f^{-1}(y)f^{-1}(y'))) \\ &= f^{-1}(y)f^{-1}(y') \end{aligned}$$

$\square$

Proposition 1.40. *The relation "is isomorphic to" is an equivalence relation on the class of groups. Its equivalence classes are called isomorphism classes.*

Proof. Induced by (1.39) and (1.38). $\square$

1.1.3 Normal subgroup and quotient

Definition 1.41. Let G be a group. A subgroup H of G is said to be a normal subgroup if :

$$\forall h \in H, \forall g \in G, ghg^{-1} \in H$$

Remark 1.42. If G is an abelian group, then any subgroup of G is a normal subgroup.

Definition 1.43. Let G be a group and H a subgroup of G . For $g \in G$, we denote the subset $gH = \{gh, h \in H\}$, called right class of g modulo H . The right classes modulo H constitute a partition of G ; The set of right classes modulo H is called right quotient of G by H , denoted by G/H ; the map

$$\begin{aligned}\pi : G &\longrightarrow G/H \\ g &\longmapsto gH\end{aligned}$$

is called canonical surjection.

Definition 1.44. Let G be a group and H a subgroup of G . The indice of H in G , denoted by $[G : H]$, is the cardinal of the quotient G/H

Proposition 1.45. *Let G be a group and H a subgroup of G . Then we have*

$$|G| = [G : H]|H|$$

Proof. Admitted. □

Remark 1.46. If the order of G is finite, then the order of H divides the order of G .